Editable_Team_Involvement_in_Demonstration_TemplateScalability & Future Plan

Date	02 July 2026
Team ID	
Project Name	Rising Waters-ML Flood Prediction System
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Uses a static dataset for prediction.	Predictions are not based on real-time weather data.	High
2	Supports a limited number of users at a time.	Performance may decrease with many users.	Medium
3	Web application is designed for desktop use only as deployed on render cloud.	Mobile accessibility is limited.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small number of users.	Deploy on cloud with auto-scaling support.
2	Data Storage	Uses a local dataset.	Integrate cloud database and real-time weather data.
3	Performance	Single machine deployment.	Optimize model and use faster cloud servers.
4	Security	Basic input validation.	Add user authentication and secure API access.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve prediction accuracy using additional weather parameters.	1 month	More accurate flood predictions.
Phase 2	Integrate real-time weather API and live rainfall data.	2 months	Real-time flood risk prediction.
Phase 3	Develop an Android mobile application.	3 months	Easy access for users on mobile devices.

Phase 4	Deploy on IBM Cloud with automated alerts via SMS and Email.	4 months	Wider accessibility and faster emergency notifications.